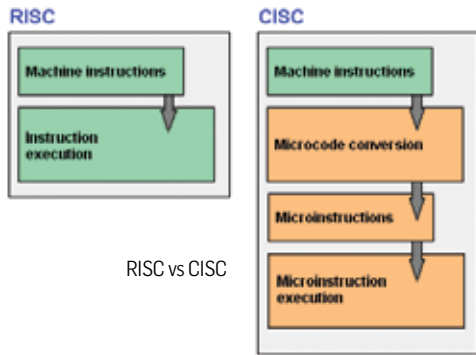


need to store all the numbers from 0 to 50 and the processor should be able to process those numbers. In such cases, an 8-bit processor (one which can process 8 bits in one clock cycle) is more than enough since it will be able to process any number between 0 and 127 (128 numbers) without having to resort to any extra tricks. Given the fact that elevator control systems only perform two types of operations – comparison and check, this processor should work fine enough.

This however may not be true for a Wi-Fi router; the reason being what it does – routing. One essential function of data routing is ensuring that the data packet received is intact. Now, most systems work with IPv4 which comes with a 16-bit header checksum to ensure data integrity.

If the processor in this case were to be 8-bit, calculating and comparing the checksum would need extra computing cycles along with some programming tricks to accomplish the task, and that would consume more power. A 16-bit processor would be better in this case. Hence, it's important that the processor in an embedded system match the word-length requirement to get the job done without needing more computing cycles, electric power or tricky programming.



Memory addressability

In most cases, the word length of a processor is the same as the number of bits that it can address on the memory bus. While this is true in most cases, there are always exceptions. One of the best examples is the PAE extension on most desktop Intel processors which can allow the processor to work with more than 4GB of memory (addressable by 32 bits) while keeping the word length as only 32 bits. With that small example, we mean to convey that the same can be the case in an embedded system as well.

An 8-bit processor might have to work with more than 256 bytes of data and hence memory addressability is another important factor. If the selected processor can't address the amount of data the program has to work upon, then the system can crash, hang, or do something that can produce catastrophic results.